# THE DROP QUESTION: A BIRD'S-EYE VIEW

Labels as in `status.pdf`. Details: `status.pdf`, `fibres.pdf`, `notes/R6/`, `notes/G-sources/`.

## 1. Set-up

For an irreducible flat connection $\alpha$ (either SO(3) bundle), $c(\alpha) = (3 + \rho(\alpha))/16$, with $\rho$ the adjoint $\rho$-invariant. This normalisation is canonical for both bundles, so differences between bundles are meaningful. $\lambda_0, \lambda_1$ are the least levels of nonzero classes for $w_2 = 0$ and $w_2 \neq 0$; $\lambda_0 = \ell$. For $\pm 2$ surgery on $K \subset S^3$:

$$u = \lambda_0(Y_2) - \lambda_0(Y_{-2}), \quad w = \lambda_1(Y_2) - \lambda_1(Y_{-2}), \quad m_1 = \lambda_0(Y_2) - \lambda_1(Y_{-2}), \quad m_2 = \lambda_1(Y_2) - \lambda_0(Y_{-2}),$$

and the both-bundle drop $D = \min(\lambda_0, \lambda_1)(Y_2) - \min(\lambda_0, \lambda_1)(Y_{-2})$.

## 2. $\pm 2$ surgery on knots in $S^3$ (computed, exact)

Exact data exist only where $Y_{\pm 2}$ are Seifert fibred, and there the differential vanishes. Among all prime knots with at most 15 crossings and the census knots, these are the torus knots and $4_1$. Data: all 210 torus knots with $pq \leq 120$ in both chiralities, plus $4_1$.

| | $u$ | $w$ | $m_1$ | $m_2$ | $D$ |
|---|---|---|---|---|---|
| positive $T(p,q)$, $n = pq$ | $-\frac{1}{8} + \frac{2n-5}{n^2-4}$, *changes sign* | $\frac{3}{8} - \frac{4}{n^2-4}$ | $(\frac{1}{8}, \frac{3}{16}]$ | $(\frac{1}{8}, \frac{3}{16}]$ | $> \frac{1}{8}$ |
| negative $T(p,q)$ | $[\frac{3}{32}, \frac{1}{8}]$ | $[0.0854, \frac{1}{8}]$ | $[0.0854, \frac{5}{32}]$ | $[\frac{1}{16}, 0.137]$ | $\geq \frac{1}{8} - \frac{1}{884}$ |
| $4_1$ | $\frac{1}{10}$ | $\frac{1}{8}$ | $\frac{13}{80}$ | $\frac{1}{16}$ | $\frac{1}{8}$ |

So in every case $m_1, m_2, D, w \geq \frac{1}{16}$; only DLME's untwisted $u$ fails. $D$ never comes near $\frac{1}{16}$: its minimum is $\frac{1}{8} - \frac{1}{884}$ at $T(3,-5)$. The minimum of $m_1$ and $w$ is $\frac{151}{1768} \approx 0.0854$, also at $T(3,-5)$, which has $\det = 1$.

*Equality* $\frac{1}{16}$. It is attained only by $m_2$, and only for $T(2,-3)$ and $4_1$. In both, the twisted minimiser of $Y_2$ is a *binary dihedral* representation: traceless meridian, longitude $\mapsto +1$, the point $(\frac{1}{4}, 0)$ of the pillowcase. That point is a common point of the two twisted surgery lines. The minimiser is absorbed into the traceless theory of $(S^3, K)$ by a flat singular connection, at zero energy, and the next twisted class is then matched with $\lambda_0(Y_{-2})$.

- For $T(2,-3)$, $m_2 = \frac{3}{16} - \frac{1}{8}$ exactly. Here $\frac{3}{16}$ is the gap at a mixed common point, namely the $\mathbb{R}P^3$ term $\frac{1}{8}$ plus the $L(4,1)$ term $\frac{1}{16}$, and $\frac{1}{8}$ is the spacing of the twisted generators.
- For $4_1$, $m_2 = \frac{13}{80} - \frac{1}{10}$. I found no structural reason for this value; it may be a coincidence.

There are $(\det K - 1)/2$ binary dihedral representations, so *none when* $\Delta_K = 1$, and this equality mechanism is absent in the case of interest. For $\det K = 1$ torus knots the smallest values are $m \approx 0.0854$ and $D = \frac{1}{8} - \frac{1}{884}$.

$\Delta_K = 1$ *knots (generator level only; computed numerically).* These are Kinoshita–Terasaka, Conway, and the Whitehead doubles of the trefoil, in both chiralities. For them the drop in the least *generator* level, over both bundles, lies in $[0.085, 0.187]$, above $\frac{1}{16}$; this is not $\ell$. For $P(-3,5,7)$ it is undecided: there are degenerate loops of traceless representations with longitude $\mapsto 1$, which are not binary dihedral.

## 3. Other surgery pairs (computed)

$-1/n$ *on torus knots in* $S^3$. The two-step drop $\ell(Y_{-1/n}) - \ell(Y_{-1/(n+2)})$ is at least $\frac{1}{8}$. Equality holds exactly for odd–odd mirror torus knots at odd $n$, where the minimiser has meridian and longitude both traceless. One-step drops have no positive floor.

$-1/n$ *on fibres of Brieskorn spheres (rechecked independently).* On the link-oriented side the two-step drop is $\frac{1}{8} + \beta^2/(8mm'')$, with equality iff the holonomy along the surgered fibre is traceless. In closed form, $\ell = A^* - \frac{m}{P}\alpha(\frac{1}{2} - \alpha)$: per step this is $\frac{1}{16}$, the value at traceless holonomy, plus a square. On the reversed side the one-sided $\ell$ rises at one step, and its two-step drops tend to 0.

## 4. The common pattern (plausible)

In all three families, *equality occurs exactly when the minimiser sits at a representation of the knot exterior that is common to both fillings*, with traceless holonomy. There a flat connection on the connecting cobordism exists, and the drop is the $\rho$-term of that flat connection at a lens-space end. For the $-1/n$ families this is the $\mathbb{R}P^3$ rotation by $\pi$, giving $\frac{1}{8}$. For $\pm 2$ the equality also involves the zero-energy absorption above, and is explained structurally only for $T(2,-3)$. In the Brieskorn chains the excess over equality is a perfect square. Every exact case is Seifert fibred.

## 5. What did not work

- *Triangles.* The exact triangles, with all levels, positivity of energy, arbitrary differentials, gradings, Euler characteristics, the mirror and cover symmetries, and the automorphisms forced by a cosmetic pair, cannot prove any drop (no-go theorem, five explicit models). The constant $\frac{1}{16}$ appears in them only as a ceiling.

- *Pillowcase steepness.* This was an energy hypothesis in disguise, and is withdrawn.
- *Equivariant theory on the branched cover.* Abandoned.
- *General homology spheres.* There is no uniform positive drop for the one-sided $\ell$; with nonzero $I(Y)$, a standard triangle argument shows no sign of working.

## 6. What this suggests

The inequality is about *where* the bottom class sits relative to common representations, and whether it survives. Equality needs a zero-energy absorption at a common point with traceless meridian and longitude $\mapsto \pm 1$. When $\Delta_K = 1$ the only candidates are degenerate traceless representations with longitude $\mapsto \pm 1$ that are not binary dihedral, as for $P(-3, 5, 7)$ and the Whitehead doubles. Any argument must either exclude these or control them. Exact evidence beyond Seifert fibred spaces is missing, and computing it needs the differential.